

Socio-Demographic and Labor Statistics Analysis of Indian States

Dataset: 2011 Census PCA Data

GROUP: 12

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Introduction

Background and Motivation

India’s socio-demographic landscape is characterized by enormous diversity across its 28 states and 7 union territories. The 2011 Census, conducted by the Office of the Registrar General and Census Commissioner of India, remains one of the most comprehensive data collection exercises, capturing population dynamics, employment patterns, literacy levels, and economic activities across the entire country.

Understanding these patterns is essential for policymakers, economists, and development researchers. This report presents a systematic analysis of the 2011 Census PCA (Primary Census Abstract) data, focusing on labor market conditions, dependency structures, agricultural engagement, and the role of literacy in shaping employment outcomes.

Research Questions

This analysis is organized around **three primary research questions**:

Research Questions

1. **RQ1 – Literacy and Employment Quality:** Are literate people more prone to get employed for periods of 6 months or more? (Answered in Analysis-4)

2. **RQ2 – Unemployment and Labor Market Status:** What is the unemployment status of states, measured using NSSO classification methods (UPSS and CWS)? (Answered in Analysis-2)

3. **RQ3 – Agricultural Dependency:** What is the agricultural dependency per state, and how are literacy and gender dynamics linked to it? (Answered in Analysis-3)

Additional Analysis

Two additional topics are included, due to their significance in understanding the other Research Questions:

Additional Topics

1. **Dependency Ratio :** Measures the economic burden on the working population. Provides important context for RQ2 and social security planning. *Not a direct research question but included due to strong analytical value.*

2. **Urbanization Drivers :** Explores the relationship between sector of employment and urbanization patterns. *Connected to RQ3 – understanding what replaces agricultural work in urbanized states.*

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Dataset Description

The dataset selected for this analysis is sourced from the **OGD Open Government Data Platform**, specifically focusing on Socio-Demographic and Labor Statistics. The original dataset has been reorganized and shortened for this project. The official metadata classifies this table (Table PCA) as “Basic Population and Socio-Economic Data” (in the official census website).

Number of Cases: Each population category is under one of these cases – age group, working-main and marginal, caste, literate, illiterate, non-working.

Number of Variables: This dataset contains 49 variables.

COLUMN NAMES:

State Code, Level, Name, TRU, No of Households, Total Population Person, Total Population Male, Total Population Female, Population in the age group 0-6 Person, Population in the age group 0-6 Male, Population in the age group 0-6 Female, Scheduled Castes population Person, Scheduled Tribes population Person, Literates Population Person, Literates Population Male, Literates Population Female, Illiterate Persons, Illiterate Male, Illiterate Female, Total Worker Population Person, Total Worker Population Male, Total Worker Population Female, Main Working Population Person, Main Working Population Male, Main Working Population Female, Main Cultivator Population Person, Main Cultivator Population Male, Main Cultivator Population Female, Main Agricultural Labourers Population Person, Main Agricultural Labourers Population Male, Main Agricultural Labourers Population Female, Main Household Industries Population Person, Main Household Industries Population Male, Main Household Industries Population Female, Main Other Workers Population Person, Main Other Workers Population Male, Main Other Workers Population Female, Marginal Worker Population Person, Marginal Worker Population Male, Marginal Worker Population Female, Marginal Worker Population 3.6 Person, Marginal Worker Population 3.6 Male, Marginal Worker Population 3.6 Female, Marginal Worker Population 0.3 Person, Marginal Worker Population 0.3 Male, Marginal Worker Population 0.3 Female, Non Working Population Person, Non Working Population Male, Non Working Population Female.

These include Population Categories – Population in the age group 0-6 Person, Literates Population Person, Illiterates Population Person, Total Worker Population Person, Main Working Population Person, Main Cultivator Population Person, Non-Working Population Person; each of these are on the basis of **STATE**, **TRU** (Total–Rural–Urban), and **Gender**. And also has State, StateCode.

Total Population (Numerical): Represents the total number of individuals in a region, further divided into male and female populations. This is the primary measure of population size.

Age Group (0–6) Population (Numerical): Captures the number of children aged 0–6 years.

Literacy count (Numerical): Includes literate and illiterate population counts.

1 Data Cleaning

The raw dataset required several cleaning steps before analysis. All steps were performed in R:

Step 1. Handle Missing Values

Replaced all empty/blank values with NA to standardize missing data representation across all 49 variables.

Step 2. Impute Outlier Values

Identified 4 values in the dataset that were erroneous or missing. Replaced each with the **local median** of that respective column to preserve distributional shape without distortion.

Step 3. Remove National-Level Aggregates

The raw dataset included rows for All-India totals (national level). These were removed using:

```
df = df[-c(1,2,3),]
```

This ensures all analysis is conducted at the **state/UT level only**.

Step 4. Derive New Analytical Columns

Added the following computed columns to support analysis:

- Dependency Ratio
- UPSS Proxy Rate
- Unemployment Proxy
- Urbanization
- Labor Distress Index
- Literacy Rate

2 Analysis: Agricultural Dependency per State

Research Question 3 (RQ3): What is the agricultural dependency per state? How are literacy and gender dynamics linked to agricultural employment? The “WHY” can be determined from Literacy Rate and Gender breakdown of each state.

People in direct involvement in agriculture:

$$\text{Agri_Workers_Person} = \text{Main Cultivator Population Person} + \text{Main Agricultural Labourers Population Person} \quad (1)$$

These are only main workers involved in agriculture (not marginal agricultural workers).

Proportion of TOTAL POPULATION directly involved in agriculture:

$$\text{Agri_Population_Share} = \frac{\text{Agri_Workers_Person}}{\text{Total Population Person}} \times 100 \quad (2)$$

Agri_Workforce_Share gives the number of people involved in agri per 100 main workers:

$$\text{Agri_Workforce_Share} = \frac{\text{Agri_Workers_Person}}{\text{Main_Workers_Person}} \times 100 \quad (3)$$

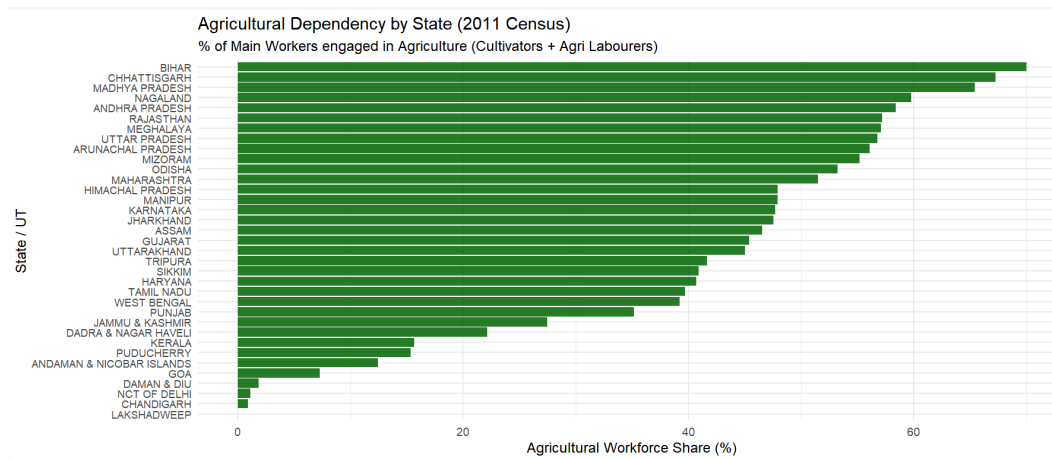


Figure 1: Agricultural Dependency by State/UT – % of Main Workers engaged in Agriculture

This is a plot for agricultural workers/cultivators relative to main workers (i.e., the agriculture share). The states at the top of the chart (Bihar, Chhattisgarh, Madhya Pradesh) rely overwhelmingly on agriculture, while urban UTs at the bottom (Chandigarh, Delhi, Daman & Diu) have near-zero agricultural engagement.

The bottom few States/UTs with lowest agricultural involvement:

State/UT	Agri_Population_Share	Agri_Workforce_Share	Literacy_Rate
GOA	2.387	7.314	88.70
DAMAN & DIU	0.880	1.838	87.10
NCT OF DELHI	0.353	1.116	86.21
CHANDIGARH	0.338	0.924	86.05
LAKSHADWEEP	0.000	0.000	91.85

Table 1: States with lowest agricultural dependency (all have high literacy, 86–92%)

A clear pattern emerges: every state with near-zero agricultural dependency also has a literacy rate above 86%. This strongly suggests that education enables occupational diversification.

2.1 Hypothesis Test: Does Higher Literacy Reduce Agricultural Dependency?

Agriculture and literacy rate are closely related. It is a common assertion that higher literate populations shift out of direct agricultural work to pursue further studies and higher-skill employment. We tested this theory:

Correlation coefficient (r) = **-0.6846**

The p -value = 0 (effectively zero), which is clear from the scatter plot below showing a strong linear model.

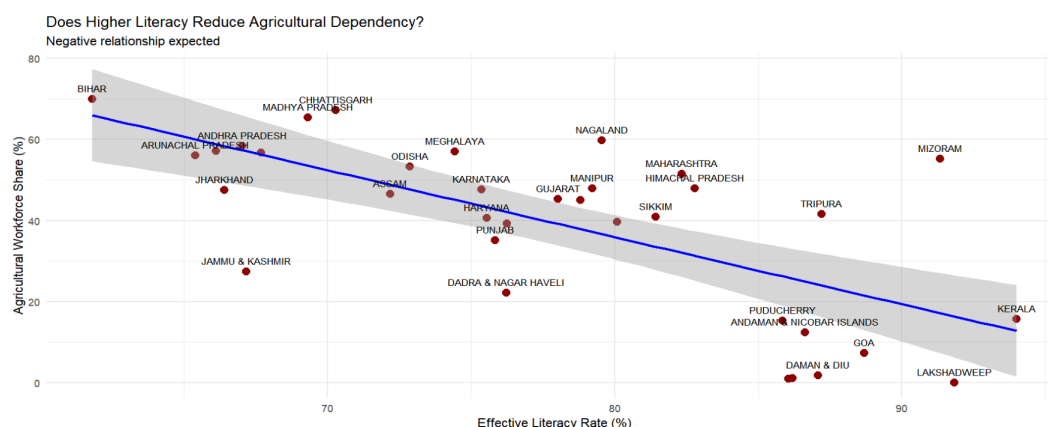


Figure 2: Does Higher Literacy Reduce Agricultural Dependency? ($r = -0.6846$, $p \approx 0$) – Negative relationship confirmed

CONCLUSION: HYPOTHESIS STRONGLY VALIDATED. Higher literacy states definitively have lower agricultural dependency. The relationship is strong ($r = -0.68$) and highly significant ($p \approx 0$).

2.2 Gender Gap in Agricultural Employment

From this plot it is clear that **Bihar** has the highest Agriculture Workforce Share and the lowest Literacy Rate among all states. More importantly, we found that the **Gender Gap** ($\text{Gender_Gap} = \text{Agri_Population_Share_Male} - \text{Agri_Population_Share_Female}$) is very high = 16.42. Meaning male-dominant agriculture share is prevalent here, with low literacy.

Same is the case with **Uttar Pradesh**: slightly higher Literacy Rate (67%), but an even larger gender gap (16.57). This is a striking finding – even with better education than Bihar, UP’s agricultural gender gap is higher, suggesting that social and cultural factors beyond literacy play a role in excluding women from the formal agricultural workforce.

3 Analysis: Literacy Rate versus Working Status

Research Question 1 (RQ1): Are literate people more prone to get employed for periods of 6 months or more? Does literacy affect the quantity of employment or the quality (stability) of employment?

Correlation between Literacy and Working Status: **-0.0062**

The scatter plot of Workers and Literacy rate shows near-zero correlation – an almost straight horizontal line – suggesting that literacy has no direct relationship with the *number* of workers in a state.

BUT if we look at main workers and marginal workers separately, we see:

Correlation	r value	Interpretation
Literacy vs Total Workers	-0.0062	Near ZERO – no quantity relationship
Literacy vs Main Workers	+0.304	Positive – literate areas have more stable jobs
Literacy vs Marginal Workers	-0.4395	Negative – illiterate areas have more unstable jobs

Table 2: Literacy correlations with different worker categories

These are significant and opposite numbers, meaning the two effects **cancel out** to give ≈ 0 correlation for the overall worker category.

This proves that **literacy in India does not have a relationship with the quantity of workers, but it has a clear relationship with the type of work.**

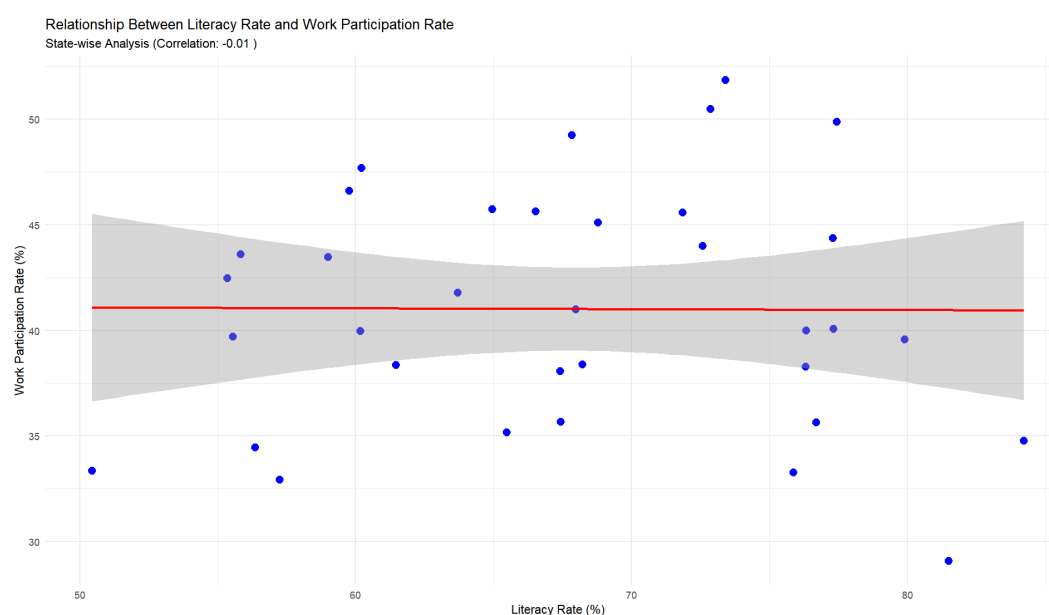


Figure 3: Relationship Between Literacy Rate and Work Participation Rate (Overall Correlation: -0.01 – near zero)

CONCLUSION: People in higher literacy categories are more prone to get employed in **Stable Jobs (Main Workers, 6+ months)**. Literacy does not create more employment – it improves the quality and stability of employment. This has direct implications for education policy: investing in literacy is an anti-precarity measure, not just an anti-unemployment measure.

4 Analysis-1: Dependency Ratio

Type: Additional topic to clearly understand RQ2

Relevance: Provides economic burden context that supports interpretation of RQ2 (employment status). High dependency ratios indicate why employment pressure and social security needs are high in certain states.

4.1 Definition

According to NSSO: A demographic indicator showing the ratio of dependents (those under 15 or over 64) compared to the working-age population (15–64). It signifies the economic burden on the productive workforce, with a lower ratio suggesting higher potential for economic growth.

$$\text{Dependency Ratio} = \frac{\text{Children}(0-14) + \text{Elderly}(65+)}{\text{Working force}(15-64)} \times 100 \quad (4)$$

Adapted Definition for This Analysis:

$$\text{Dependency Ratio} = \frac{\text{Non Working Population Person}}{\text{Total Worker Population Person}} \times 100 \quad (5)$$

This is the standard economic measure used for:

- Social security planning
- Labour force burden assessment
- Economic productivity analysis

Why this works: Census does not have the 0–14 / 15–64 / 65+ split. Our version uses Non-Working population as a practical proxy for all dependents (children 0–6 + elderly + unemployed + students + homemakers). The ratio tells us: *how many non-workers must be supported by each 100 workers.*

4.2 Dependency Ratio by State

	Name	TRU Total Population Person	Population in the age group 0-6 Person		
1	JAMMU & KASHMIR Total	12541302	2018905		
2	HIMACHAL PRADESH Total	9108060	1593008		
3	PUNJAB Total	3433242	425897		
4	CHANDIGARH Total	6864602	777898		
5	UTTARAKHAND Total	6176050	712822		
6	HARYANA Total	688552	65076		
	Non Working Population Person	Total Worker Population Person	Dependency Ratio	Young Dependency Ratio	Dependency Ratio
1	8218589	4322713	190.12572		46.70458
2	5994979	3113081	192.57382		51.17143
3	2223610	1209632	183.82533		35.20881
4	3305180	3559422	92.85721		21.85462
5	2886666	3289384	87.75704		21.67038
6	418514	270038	154.98337		24.09883
	Other Dependents				
1	143.42113				
2	141.40239				
3	148.61652				
4	71.00260				
5	66.08666				
6	130.88454				

Figure 4: Dataset snapshot after adding dependency ratio columns

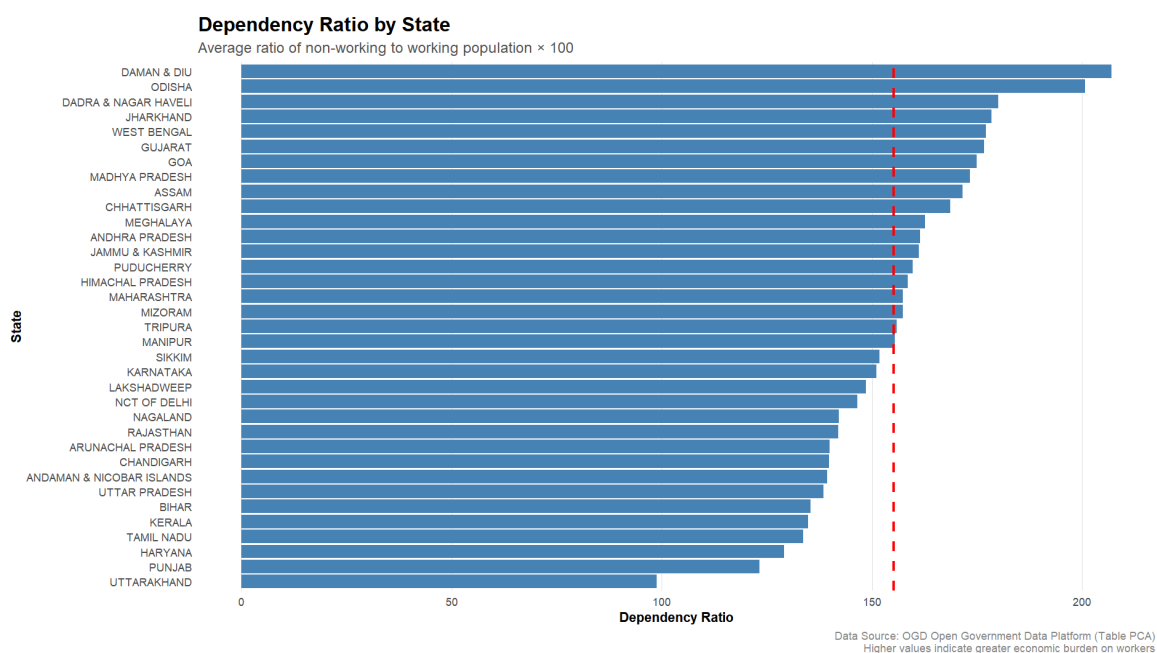


Figure 5: Dependency Ratio by State – red line = national average (155.19)

National Average Dependency Ratio = 155.1858 (RED LINE)

Summary Statistics: Min = 85.11 1st Qu = 125.44 Median = 152.67 Mean = 155.19 3rd Qu = 180.66 Max = 256.96
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Observations: Almost half of the states are above the national average in dependency. The national average itself (155) is high – meaning on average, 155 non-workers depend on every 100 workers. This signals significant economic pressure at the national level.

4.3 Dependency Ratio Heatmap by TRU (Total / Rural / Urban)

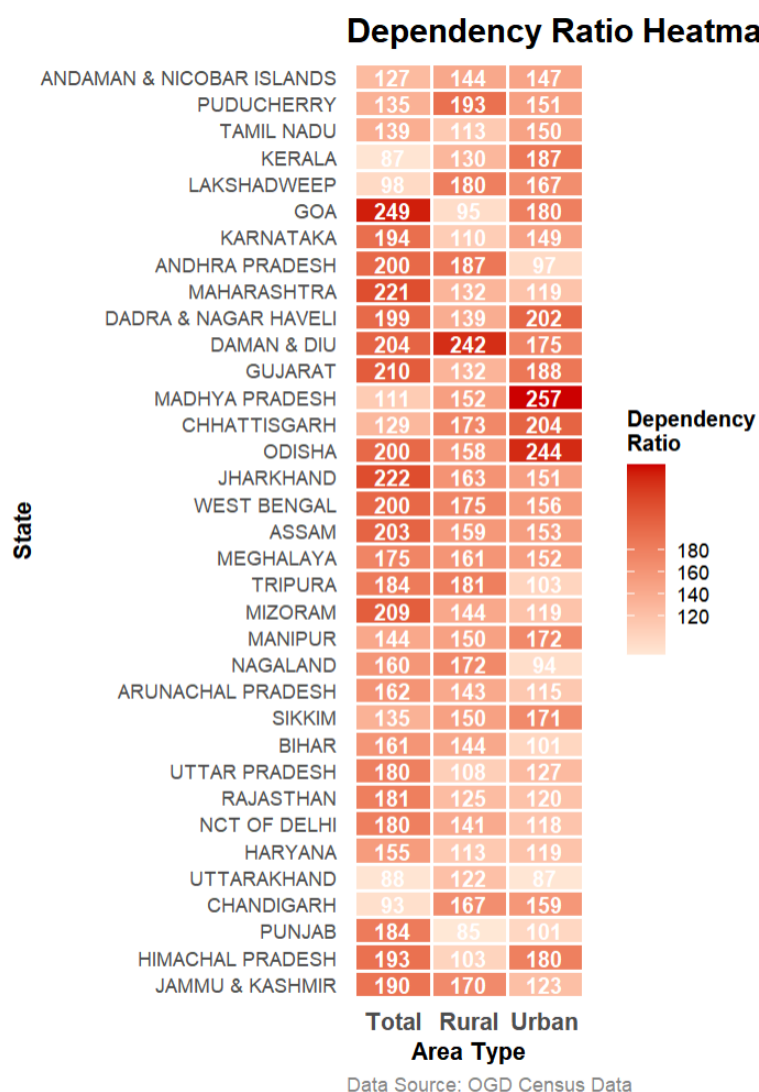


Figure 6: Dependency Ratio Heatmap × Area Type (Total, Rural, Urban)

Area Type	Mean Ratio	Median Ratio	SD
Total	169	180	41.6
Rural	147	144	31.8
Urban	150	151	40.9

Table 3: Summary statistics of dependency ratio by area type

Rural areas show lower mean dependency (147) but higher variability in some states like Madhya Pradesh and Jharkhand (visible as dark red in the heatmap). Urban areas are more consistent (SD = 40.9, median = 151).

4.4 Top 5 and Lowest 5 States

Highest Dependency Ratio		Lowest Dependency Ratio	
State	Ratio	State	Ratio
DAMAN & DIU	207	UTTARAKHAND	98.7
ODISHA	201	PUNJAB	123
DADRA & NAGAR HAVELI	180	HARYANA	129
JHARKHAND	178	TAMIL NADU	134
WEST BENGAL	177	KERALA	135

Table 4: Top 5 and Lowest 5 states by dependency ratio

4.5 Why the Difference? Root Causes

4.5.1 Reasons for high dependency

1. HIGH YOUNG POPULATION (0-6 age group):

Young population proportion – ODISHA: 37%, JHARKHAND: 40%

- Many children (need care, education)
- Parents can't all work (childcare needed)
- Example: Jharkhand has high fertility rate

2. HIGH UNEMPLOYMENT / NON-WORKING:

Jharkhand has 60% of population in non-working status, Odisha 58%, Dadra-Nagar-Haveli 54%.

- Low education → fewer job skills
- Agricultural seasonality → irregular work
- Limited industries → fewer jobs
- Example: Odisha economy is agriculture-dependent

3. AGING ISSUES:

- Elderly with no pensions
- Need healthcare support

4. MIGRATION:

- Young workers leave for cities
- Leaves behind children and elderly

4.5.2 Reasons for low dependency (Uttarakhand, Punjab, Haryana)

1. LOWER YOUNG POPULATION:

Punjab: 31%, Uttarakhand: 32% (significantly lower than crisis states)

- Fewer children per family
- Better family planning

2. HIGHER EMPLOYMENT:

- Better education → more jobs
- More industries

3. MIGRATION INFLOW:

- Young workers migrate IN
- Increases worker pool

4. BETTER EDUCATION:

- Higher literacy → better employment

5 Analysis-2: Workforce Participation

Research Question 2 (RQ2): What is the unemployment status of states? Using NSSO classification methods – Usual Principal and Subsidiary Status (UPSS) and Current Weekly/Monthly Status (CWS) – how much employment precarity exists across Indian states?

5.1 What is UPSS?

UPSS = Usual Principal + Subsidiary Status

This is an official NSSO (National Sample Survey Office) classification used in Indian labor statistics.

Status	Work Duration	Examples
Principal Status	6+ months per year	Regular jobs, continuous work
Subsidiary Status	30+ days in remaining months	Secondary part-time jobs
Marginal Status	< 3 months total	Occasional/seasonal work

Table 5: The 3 categories in UPSS

NSSO uses this to measure **real employment** because:

- Census counts everyone who worked even 1 day = Misleading
- UPSS focuses on **meaningful employment** (6+ months or regular pattern)
- Separates stable workers from occasional workers
- Standard for policy comparisons across India

5.2 Our Proxy Metrics

5.2.1 Using UPSS PROXY

Dataset doesn't have exact NSSO categories, so we create a **proxy**:

$$\text{UPSS_Proxy} = \text{Main Workers} + \text{Marginal 3-6 Workers} \quad (6)$$

What we are measuring: "Workers with some regular employment pattern"

Component	Duration	Interpretation
Main Workers	6+ months	Definitely employed (stable)
Marginal 3-6	3-6 months	Semi-employed (half-year pattern)
UPSS Proxy Total	3+ months	"Regular enough" to count as employed

EXAMPLE to understand the proxy:

Group	Count	Metric	Meaning
Main Workers (6+ months)	60	Principal Status Rate: 60%	60% have stable year-round jobs
Marginal 3-6 Workers	20	UPSS Proxy Rate: 80%	80% have work for 3+ months/year
Marginal 0-3 Workers	15	Marginal 0-3 Rate: 15%	15% work less than 3 months
Non-Workers	5	True Unemployment: 5%	5% have zero work
Total	100		

Table 6: Example breakdown of 100 people using our proxy metrics

5.2.2 Using Unemployment PROXY

Official NSSO Unemployment Rate (UPSS):

$$\text{Unemployment Rate} = \frac{\text{Number of Unemployed Persons}}{\text{Labour Force}} \times 100 \quad (7)$$

- Labour Force = Employed (Principal Status + Subsidiary Status) + Unemployed
- Unemployed = Persons who were seeking/available for work but did not work even for 30 days in the year (subsidiary status criterion) + those unemployed on principal status

What we used:

$$\text{Unemployment_Proxy} = \frac{\text{Non Working Population Person} - \text{Population in 0-6 age group}}{\text{Total Worker Population Person}} \times 100 \quad (8)$$

NSSO measures underemployment in two ways:

- Proportion of usual-status workers who were not employed in the reference week – CWS (Current Weekly Status underemployment)
- Person-days not utilised for work among workers – CDS (Current Daily Status), used to measure unemployment as well

5.2.3 Using LABOR DISTRESS INDEX

$$\text{Labor Distress Index} = \frac{\text{Marginal Worker Population 0.3} + \text{Marginal Worker Population 3.6}}{\text{Main Workers}} \times 100 \quad (9)$$

THEORETICAL BASIS:

Source 1: NSSO Employment Classification Framework

The National Sample Survey Office (NSSO) classifies workers by:

- Usual Principal Status (6+ months)
- Subsidiary Status (30+ days in remaining months)
- Marginal Status (< 3 months total)

This is the **OFFICIAL** Indian labor classification. Reference: NSSO 68th Round (2012), Ministry of Statistics & Programme Implementation.

Source 2: ILO (International Labor Organization) Standards

ILO defines “precarious employment” as work with:

- Low wages
- Unstable/temporary nature
- No benefits/protections

Marginal workers 0-3 + 3-6 fit this definition perfectly.

5.3 Workforce Participation Rates by State

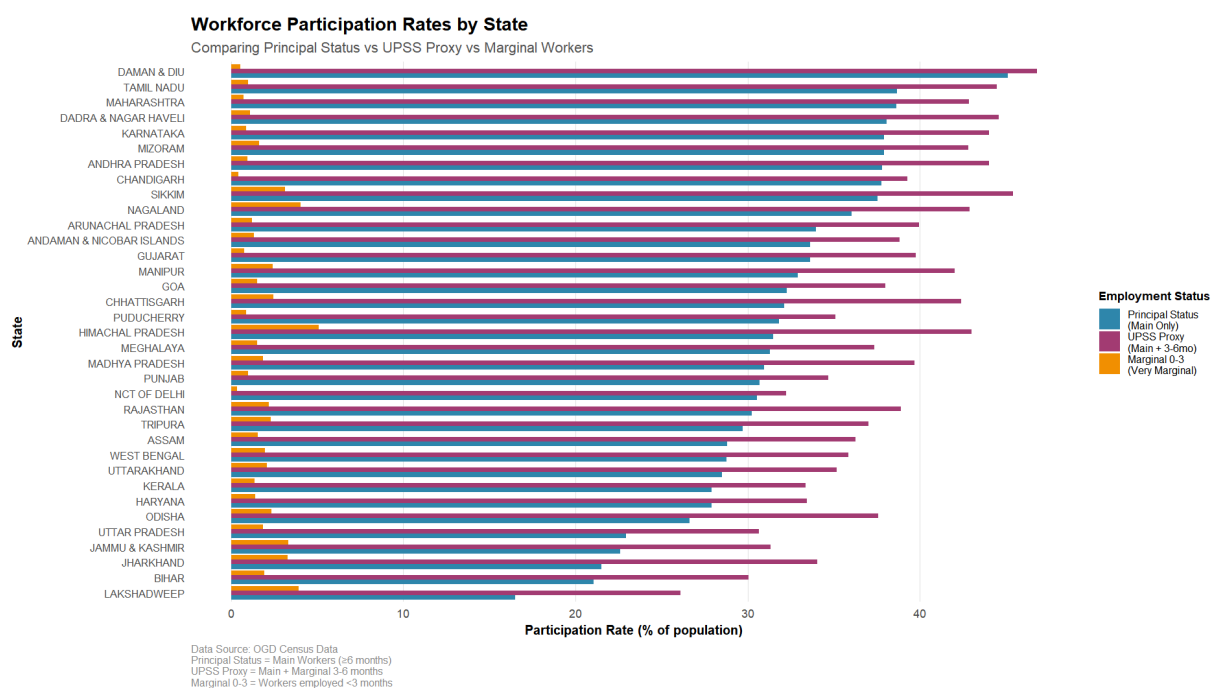


Figure 7: Workforce Participation Rates by State – Comparing Principal Status vs UPSS Proxy vs Marginal 0-3 Workers

From this plot we can see that the difference in UPSS workers and Marginal_0-3 workers is very significant for every state. Only Sikkim, Nagaland, Himachal Pradesh, and Lakshadweep have lower differences, meaning marginal workers are far more in comparison to other states.

5.4 Correlation: Validating the Labor Distress Metric

Does Labor Distress correlate with Principal Status Rate?

Correlation: **-0.7025** – As distress increases, principal status **DECREASES**

Does Labor Distress correlate with Marginal 0-3 Rate?

Correlation: **0.9014** – As distress increases, marginal workers **INCREASE**

Basically, low distress states have higher principal status workers/main workers. To see this clearly:

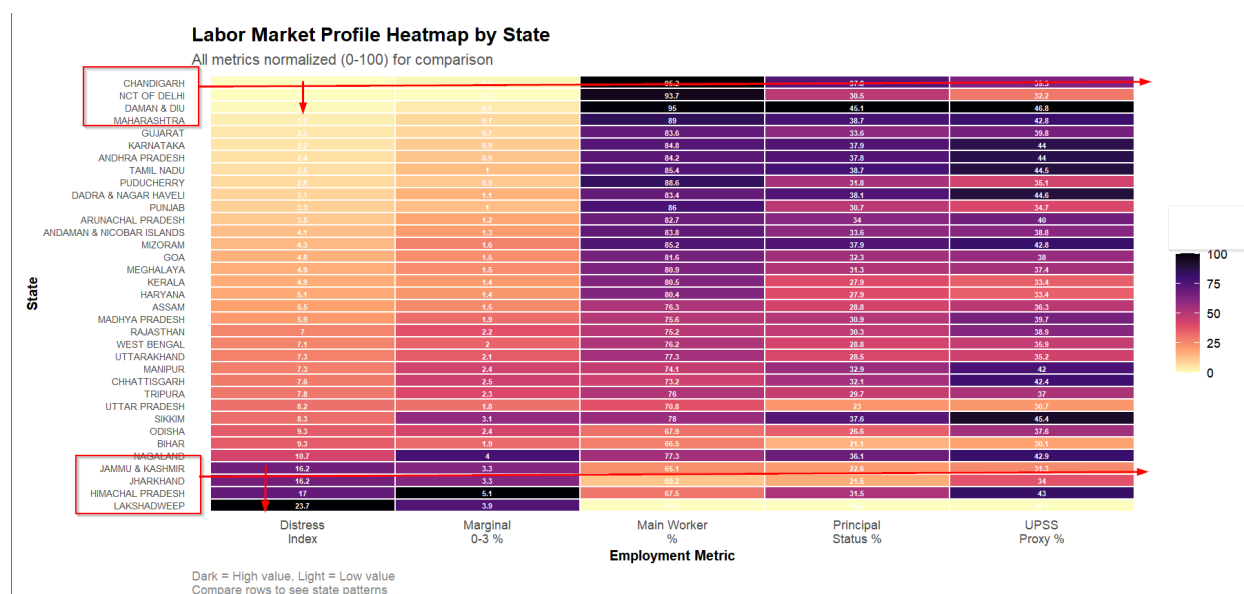


Figure 8: Increasing Distress Index and its relation to different categories of workers

5.5 Critical States: Highest Labor Distress

By comparing the labour distress index of different states and the workforce participation rates as we derived, we get the TOP 3 states that are in critical status so far as employment rate and labor distress is concerned:

5.5.1 Rank 1: LAKSHADWEEP

Labor Distress Index: 23.66

KEY METRICS:

- Main workers (% of total workers): 55.4%
- Principal Status Rate (% of population): 16.51%
- Marginal 0-3 Rate (% of population): 3.91%
- UPSS Proxy Rate (% of population): 26.09%

Conclusion:

CRISIS LEVEL – Extremely high employment precarity. 55.4% have stable jobs. 3.91% in marginal employment. Very low stable employment base, **BUT literacy rate is significantly high (91.85%)**. Hence many literate people did not have stable jobs – education alone is not sufficient without matching job creation.

THE PROBLEM:

- Widespread employment instability across the state
- Majority of workers lack secure, long-term employment
- High proportion of population dependent on irregular work
- Limited formal sector jobs
- Vulnerable to economic shocks and seasonal fluctuations

5.5.2 Rank 2: HIMACHAL PRADESH

Labor Distress Index: 17.0

KEY METRICS:

- Main workers (% of total workers): 67.5%
- Principal Status Rate (% of population): 31.49%
- Marginal 0-3 Rate (% of population): 5.1%
- UPSS Proxy Rate (% of population): 43.01%

Conclusion:

CRISIS LEVEL – Extremely high employment precarity. 67.5% have stable jobs. 5.1% of population in highly marginal employment (<3 months). Reasonable stable employment base. EVEN though the literacy rate (86%) is very high, this is the situation – suggesting Himachal Pradesh's economy is heavily tourism and agriculture-dependent, both of which are seasonal.

THE PROBLEM:

- Widespread employment instability across the state
- Majority of workers lack secure, long-term employment
- High proportion of population dependent on irregular work
- Limited formal sector jobs

5.5.3 Rank 3: JHARKHAND

Labor Distress Index: 16.22

KEY METRICS:

- Main workers (% of total workers): 60.2%
- Principal Status Rate (% of population): 21.51%
- Marginal 0-3 Rate (% of population): 3.29%
- UPSS Proxy Rate (% of population): 34.03%

Conclusion:

CRISIS LEVEL – Extremely high employment precarity. 60.2% have stable jobs. 3.29% in marginal employment. Below-average stable employment. Despite being a mineral-rich state, formal employment remains low, suggesting extractive industries have not created broad-based stable employment.

The problems faced are similar when it comes to employment and distress.

5.6 Healthy States: Lowest Labor Distress

5.6.1 Rank 1: CHANDIGARH

Labor Distress Index: 1.07

KEY METRICS:

- Main Workers (% of total workers): 95.2%
- Principal Status Rate (% of population): 37.77%
- Marginal 0-3 Rate (% of population): 0.41%

- UPSS Proxy Rate (% of population): 39.27%

Conclusion:

HEALTHY – Low employment precarity. Strong majority (95.2%) have stable jobs. Very low marginal employment (< 0.41%). Strong stable employment base.

STRENGTHS:

- Stable labor market with secure employment
- Low unemployment and underemployment
- Strong formal sector presence (government, services)
- Less vulnerable to economic shocks
- Better income security for workers

5.6.2 Rank 2: DAMAN & DIU

Labor Distress Index: 1.25

KEY METRICS:

- Main Workers (% of total workers): 95%
- Principal Status Rate (% of population): 45.1%
- Marginal 0-3 Rate (% of population): 0.54%
- UPSS Proxy Rate (% of population): 46.8%

Conclusion:

HEALTHY – Low employment precarity.

Strong majority (95%) have stable jobs.

Very low marginal employment (< 0.54%).

Highest principal status rate in the nation (45.1%) – indicating a strongly industrialized formal economy.

5.7 Hypothesis Test: Distress Index vs Literacy Rate

Statistical Validation of Labor Distress Metric. States with higher Distress Index should have lower literacy.

$$H_0 : \rho = 0 \quad (\text{No correlation between Distress Index and Literacy Rate}) \quad (10)$$

$$H_1 : \rho < 0 \quad (\text{Negative correlation: Higher distress} \rightarrow \text{Lower literacy}) \quad (11)$$

SIGNIFICANCE LEVEL (α): $\alpha = 0.05$ (95% confidence level)

EXPECTED OUTCOME: Correlation coefficient < -0.6 (Strong negative); p -value < 0.05

FINDINGS:

Correlation (r) = -0.4035 – **WEAK** correlation, 16.28% of variance explained

P -Value = 0.0081 ($p < 0.01$) – **VERY SIGNIFICANT**

Confidence interval: $[-1, -0.1362]$

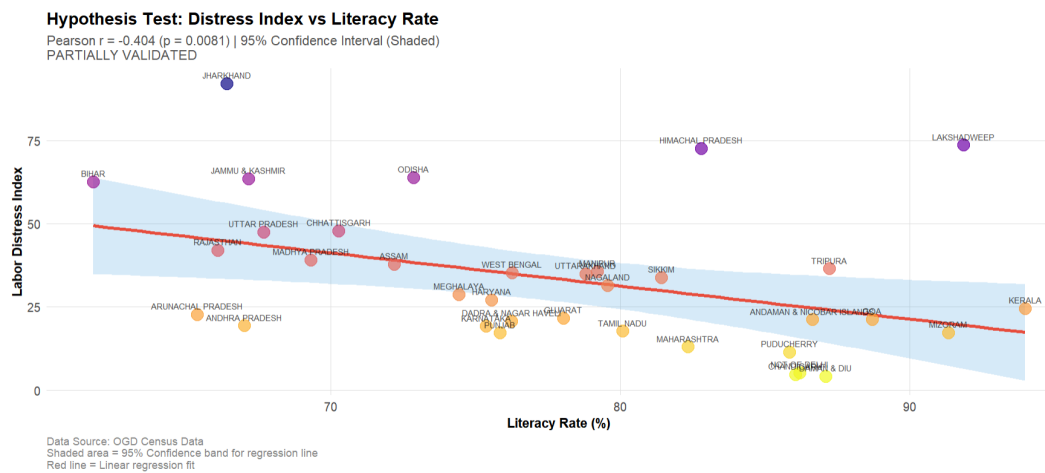


Figure 9: Hypothesis Test: Distress Index vs Literacy Rate (Pearson $r = -0.4035$, $p = 0.0081$, partially validated)

CONCLUSION: HYPOTHESIS PARTIALLY VALIDATED

Relationship is statistically significant ($p < 0.05$). But correlation strength is 0.40, not < -0.6 as expected. Still shows meaningful relationship, just weaker than expected. This confirms that literacy and labor precarity are related – but literacy alone does not fully explain distress levels; economic structure, industrialization, and geographic factors also play a role.

6 Analysis (Part 2): Urbanization and Sector Dynamics

Type: Additional Topic

Connection to Research Questions: Directly connected to **RQ3** (Agricultural Dependency). Understanding which sectors drive urbanization helps explain what replaces agricultural work in developed states – answering the “why” of occupational transition from agriculture to services.

To understand the driving forces behind urbanization for any State/UT, we need to look at where its working population is concentrated, specifically the Household Industries and Service sector. Agriculture and cultivation generally do not contribute to urbanization (by intuition), which will be proved by the data below.

6.1 Findings

First we formulated the definition of Urbanization:

$$\text{Urbanization} = \frac{\text{Urban Population}}{\text{Total Population Person}} \times 100 \quad (12)$$

- **Negative Correlation between Household Industries and Other Workers (-0.367).** This shows that states tend to specialize in either traditional household industries OR modern non-agricultural sectors – not both at high levels simultaneously. This reflects India’s dual economy structure.
- **Main Other Workers Share has a Strong Positive Correlation (+0.816) with Urbanization.** Main Other Workers includes: Manufacturing, Trade, Hotels & Restaurants, Transport & Communication, Banking, Insurance, Real Estate, Public Administration, Education, Health Services, etc.
- **Main Household Industries Share has a Negative Correlation (-0.299) with Urbanization.** Household Industries include: Handloom, Handicrafts, Pottery, Weaving, Small Cottage Industries, etc. These are traditional, rural-based activities – so this outcome is expected and valid.

It is clear that the **service sector is more correlated with urbanization** than household industries are. This is the plot that clearly justifies the findings:

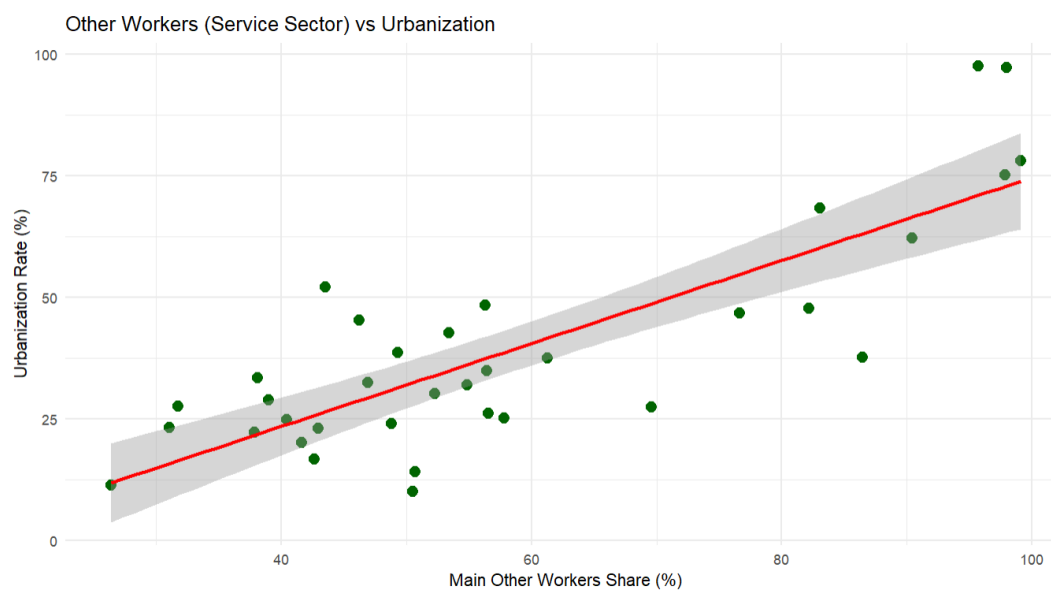


Figure 10: Urbanization correlation with Other Workers (Service sectors) – strong positive relationship ($r = +0.816$)

Interpretation: States with a higher proportion of workers in the service sector are significantly more urbanized. This makes intuitive sense – service sector jobs (government, banking, education, health) are concentrated in cities. This also means that states looking to urbanize and reduce agricultural dependency must build service sector capacity, not just move away from household industries.

7 Analysis (Part 3): Urbanization and Agriculture Workforce Share

Type: Additional Topic

Connection to above analysis: Directly connected to **Urbanization vs Service sector relation** based on RQ3 (Agricultural Dependency). This explains the exact mirror image of the service sectors and its urbanizing factor.

Just as we saw the strong positive correlation between Service sectors and urbanization, we tried the same for Agricultural sectors as well

7.1 Findings

- **The relationship is strong ($r = -0.814$):**

States with low agricultural share (0-10%) have high urbanization (75-97%) — these are the top-left cluster (Delhi, Chandigarh, Lakshadweep, Daman Diu) States with high agricultural share (60-70%) have low urbanization (10-25%) — bottom-right (Bihar, Jharkhand, Chhattisgarh) This shows that states tend to specialize in either traditional household industries OR modern non-agricultural sectors – not both at high levels simultaneously. This reflects India's dual economy structure.

- The point near (0, 97%) is Lakshadweep and Delhi — near-zero agriculture but maximum urbanization Some points around (40-60%, 45-52%) break the trend slightly — these are Mizoram and Nagaland which have high agriculture but moderate urbanization due to geographic/tribal factors
- **What this means economically:** Agriculture and urbanization are structurally competing forces — as agricultural share rises, urban economy shrinks Correlation is almost equal to the service sector correlation (+0.816 vs -0.814) — meaning both sides of the same story are equally strong

This is the plot that clearly justifies the findings and easily comparable to above one:

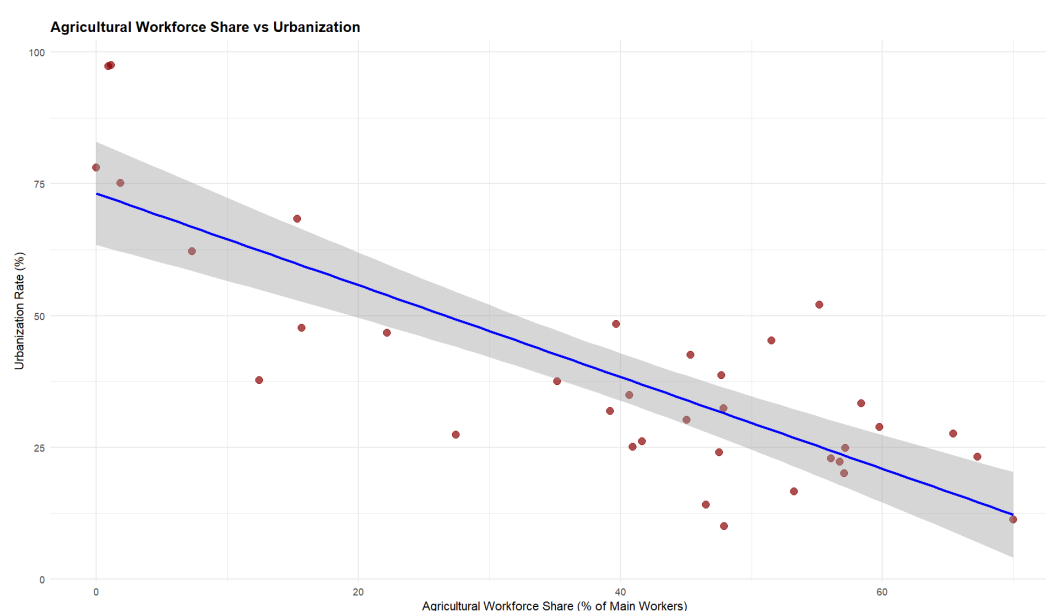


Figure 11: Urbanization correlation with Agricultural Workers Share – strong negative relationship ($r = +0.814$)

Conclusion

This report presented a multi-dimensional analysis of India's labor market and socio-demographic structure using 2011 Census PCA data across 35 states and union territories. The findings are summarized below.

RQ1: Literacy and Employment Quality

Literacy does not increase the overall quantity of employment ($r \approx 0$), but it significantly improves employment quality. Literate areas have more main workers (stable, 6+ month jobs, $r = +0.304$), and illiterate areas have more marginal workers (precarious work, $r = -0.44$). Education policy should be seen as a tool for employment stabilization, not just workforce expansion.

RQ2: Unemployment and Labor Market Status

The Labor Distress Index (validated with strong correlations: -0.70 with principal status, $+0.90$ with marginal 0-3) reveals severe employment precarity in Lakshadweep (distress = 23.66), Himachal Pradesh (17.0), and Jharkhand (16.22). Chandigarh and Daman & Diu represent the healthiest labor markets. A stark rural-urban divide exists: rural distress (9.19) is 3x higher than urban (3.08). The hypothesis linking distress to literacy was partially validated ($r = -0.40$, $p = 0.008$), confirming that economic structure matters beyond just education.

RQ3: Agricultural Dependency

Higher literacy strongly reduces agricultural dependency ($r = -0.68$, $p \approx 0$). States with the lowest agricultural engagement (Chandigarh, Delhi, Lakshadweep) all have literacy rates above 86%. Critically, gender gaps in agricultural employment persist even in moderately literate states – UP's gender gap (16.57) exceeds Bihar's (16.42) despite higher literacy, pointing to entrenched social factors beyond education.

Additional Findings

The dependency ratio analysis (Analysis 1) shows that national economic burden on workers is significantly high (national average = 155). Urbanization analysis (Analysis 2, Part 2) confirms that the service sector ($r = +0.816$) is the primary driver of urbanization – states must build service sector capacity to successfully transition from agricultural economies.

Overall: India's labor market challenges are deeply interconnected. Low literacy $\rightarrow$ high agricultural dependency $\rightarrow$ seasonal/precarious work $\rightarrow$ high distress $\rightarrow$ high dependency ratios. Breaking this cycle requires simultaneous investment in education, industrial development, and gender-inclusive policies.

References

- [1] National Sample Survey Office (NSSO), *Employment and Unemployment Survey: Key Indicators of Employment and Unemployment in India*, 68th Round (2011-12), Ministry of Statistics & Programme Implementation, Government of India.
- [2] Office of the Registrar General & Census Commissioner, India, *Primary Census Abstract (PCA) – Table PCA*, Census of India 2011, OGD Open Government Data Platform.
- [3] International Labour Organization (ILO), *Part-time work: Solution or trap?*, ILO Conditions of Work and Welfare Facilities Branch. Geneva: ILO, 1997. (ILO Convention 175 on Part-Time Work.)
- [4] Mehrotra, S. and Parida, J.K. (2017), *Why is the Labour Force Participation of Women Declining in India?*, World Development, Vol. 98, pp. 360–380.
- [5] Dreze, J. and Sen, A. (2002), *India: Development and Participation*, Oxford University Press, New Delhi.
- [6] World Bank (2013), *India – Poverty Assessment: Toward More Inclusive Growth*, World Bank Report No. 70296-IN.